Univariate Distributions: 20 Worked Problems

Binomial · Geometric/NegBin · Poisson · Uniform · Exponential · Normal

Six distribution families, each color-coded so you can find "the Poisson box" at a glance during review. Every problem carries a plain-language analogy, because these six distributions aren't six unrelated formulas — they're one family tree (Binomial → Poisson as $n \to \infty$; Geometric → Exponential as the continuous limit; sums and races of Exponentials → Gamma/harmonic numbers; sums of many small effects → Normal), and the analogies below are chosen to make those connections visible.

A. Binomial Distribution

Problem A1 — Moments and the mode

A market-making algorithm sends 20 independent quotes, each filled with probability $p = 0.3$. Find the mean, variance, and the *mode* (most likely fill count).

Solution. $\mu = np = 20(0.3) = \boxed{6}$. $\sigma^2 = np(1-p) = 20(0.3)(0.7) = \boxed{4.2}$. **Mode:** for $\text{Bin}(n,p)$, the mode is $\lfloor (n+1)p \rfloor$ (with a tie at $(n+1)p-1$ if $(n+1)p$ is itself an integer — see Problem A3). Here $(n+1)p = 21(0.3) = 6.3$, so the mode is $\lfloor 6.3 \rfloor = \boxed{6}$.

★ *Intuition:* The mean is "what you'd average over many days"; the mode is "what you'll most likely actually see today" — for Binomial these are close but not identical, since the distribution is slightly skewed unless $p = \frac{1}{2}$.

Problem A2 — Poisson approximation to Binomial

An options book has 10,000 contracts, each independently carrying a 0.0003 probability of a pin-risk event. Approximate $P(\text{at least 2 pin-risk events})$ using $\lambda = np$.

Solution. $\lambda = 10000(0.0003) = 3$. Using Pois(3):

$$P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-3} - 3e^{-3}$$

$$= 1 - 4e^{-3} \approx \boxed{0.8009}.$$

★ *Intuition:* This is the "Law of Rare Events" in action: 10,000 nearly-never-fires events, each individually negligible, sum into a genuinely substantial 80% chance of 2-or-more. No single contract worries you; the *portfolio* of 10,000 of them absolutely should.

▶ *Speed trick:* Whenever n is large (say ≥ 50) and p is small (say ≤ 0.05), skip the Binomial pmf entirely and go straight to Pois(np) — see A4 for exactly how good this approximation is.

Problem A3 — The mode can tie

9 independent trades each have win probability $p = 0.5$. Show the mode of the win-count is *not unique*, and find both values.

Solution. $(n+1)p = 10(0.5) = 5$, an integer — this is precisely the tie condition. Both $k = 5$ and $k = 4$ are modes:

$$\binom{9}{4} = 126, \qquad \binom{9}{5} = 126$$

$$\implies \boxed{\text{modes at } k = 4 \text{ and } k = 5}.$$

★ *Intuition:* Picture the Binomial pmf as a skyline: usually it has one tallest building. But when $(n+1)p$ lands exactly on an integer, two adjacent buildings tie for tallest — a measure-zero coincidence in the parameter space, but a common exam trap precisely because $p = 0.5$ (the most "natural" value to test) triggers it constantly.

Problem A4 — When is the Poisson approximation actually good?

Compare exact $P(X = 0)$ to the Poisson approximation $e^{-\lambda}$ for (a) n=100, p=0.03 and (b) n=20, p=0.15 — both have $\lambda = np = 3$. Which approximation is better, and why?

Solution. (a) $0.97^{100} \approx 0.04755$ vs. $e^{-3} \approx 0.04979$: relative error $\approx \boxed{4.7\%}$. (b) $0.85^{20} \approx 0.03876$ vs. $e^{-3} \approx 0.04979$: relative error $\approx \boxed{28.5\%}$ — much worse!

★ *Intuition:* Matching $\lambda = np$ is *not enough* to guarantee a good approximation — what actually matters is p being small *on its own*, independent of n. Think of it as: the Poisson limit is a statement about each individual trial becoming negligible while the *count* of trials grows to compensate; $p = 0.15$ per trial is simply not "rare" no matter how you dress up λ.

▶ *Speed trick:* Rule of thumb for Stage-1 speed: trust the Poisson approximation when $n \geq 20$ *and* $p \leq 0.05$. If p is more like 0.1–0.2, flag it as approximate at best.

B. Geometric & Negative Binomial

Problem B1 — Geometric waiting time & memorylessness

A trader attempts a specific strategy entry each day; each day independently succeeds with probability $p = 0.2$. Let T = day of first success. Find $\mathbb{E}[T]$, $\text{Var}(T)$, $P(T > 10)$, and verify $P(T > 15 \mid T > 5) = P(T > 10)$.

Solution. $\mathbb{E}[T] = \dfrac{1}{p} = \boxed{5}$. $\text{Var}(T) = \dfrac{1-p}{p^2} = \dfrac{0.8}{0.04} = \boxed{20}$.

$$P(T > 10) = (1-p)^{10} = 0.8^{10} \approx \boxed{0.1074}.$$

Memoryless check: $P(T > 15 \mid T > 5) = \dfrac{P(T > 15)}{P(T > 5)} = \dfrac{0.8^{15}}{0.8^5} = 0.8^{10} = 0.1074$. ✓ Matches $P(T > 10)$ exactly.

★ *Intuition:* The strategy has no memory of its 5 straight failures — "5 days of bad luck" doesn't make day 6 any more likely to succeed, nor less. This feels wrong to most people (the "gambler's fallacy" lives exactly here) but is a direct, provable consequence of independent daily trials.

Problem B2 — Negative Binomial: waiting for the rth success

A quant needs 4 backtests to pass a strict threshold; each independently passes with probability $p = 0.25$. Let N = number of backtests needed to get the 4th pass. Find $\mathbb{E}[N]$, $\text{Var}(N)$, and $P(N = 10)$.

Solution. $\mathbb{E}[N] = \dfrac{r}{p} = \dfrac{4}{0.25} = \boxed{16}$. $\text{Var}(N) = \dfrac{r(1-p)}{p^2} = \dfrac{4(0.75)}{0.0625} = \boxed{48}$.

$$P(N{=}10) = \binom{9}{3}(0.25)^4(0.75)^6 \approx \boxed{0.0584}.$$

★ *Intuition:* Negative Binomial answers "how long until my rth win," as opposed to Binomial's "how many wins in n fixed tries" — same ingredients (p, successes), but one fixes the trial count and asks about successes, the other fixes the success count and asks about trials. Confusing the two is a very common mislabeling error.

Problem B3 — The decomposition trick: NegBin as a sum of Geometrics

Re-derive B2's $\mathbb{E}[N]$ and $\mathrm{Var}(N)$ *without* the NegBin formulas, by writing N as a sum of i.i.d. Geometric gaps.

Solution. Let G_i = number of trials between the $(i-1)$th and ith success (so $G_1, \ldots, G_4 \overset{\text{iid}}{\sim} \text{Geom}(p)$, and $N = G_1 + G_2 + G_3 + G_4$). By linearity (valid regardless of dependence) and independence (for variance):

$$\mathbb{E}[N] = 4\,\mathbb{E}[G] = 4\left(\frac{1}{0.25}\right) = 4(4) = \boxed{16}.$$

$$\mathrm{Var}(N) = 4\,\mathrm{Var}(G) = 4\left(\frac{0.75}{0.0625}\right) = 4(12) = \boxed{48}.$$

Both match B2 exactly.

★ *Intuition:* This is the same "break a hard waiting-time problem into a sum of easy i.i.d. waiting-time problems" trick that makes the Coupon Collector problem (Notebook 1) tractable — recognizing a complicated random variable as a sum of simple, independent pieces is one of the highest-leverage moves in all of probability.

▶ *Speed trick:* Whenever you see "time until the rth occurrence of something memoryless," immediately reach for this decomposition instead of memorizing the NegBin moment formulas separately — one trick, infinitely reusable.

C. Poisson Distribution

Problem C1 — Basic Poisson & the Law of Rare Events

Large orders arrive at an RFQ desk at an average rate of $\lambda = 4$ per minute (Poisson). Find $P(X{=}2)$ and $P(X \geq 1)$ in a given minute.

Solution.

$$P(X{=}2) = \frac{e^{-4}4^2}{2!} = 8e^{-4} \approx \boxed{0.1465}.$$

$$P(X \geq 1) = 1 - P(X{=}0) = 1 - e^{-4} \approx \boxed{0.9817}.$$

★ *Intuition:* Poisson is the limit of Binomial as you chop a minute into infinitely many infinitesimal sub-intervals, each with an infinitesimal, independent chance of "one order arriving" — the Law of Rare Events says that in this limit, only the *product* $np \to \lambda$ survives; everything else about how finely you chopped time washes out.

Problem C2 — Sums of independent Poissons

Desk A receives orders at rate 3/min, Desk B at rate 5/min, independently, both Poisson. Find the distribution of the combined total, and $P(\text{total} \geq 10)$ in a minute.

Solution. Sums of independent Poissons are Poisson with summed rates: Total $\sim$ Pois(3+5) = Pois(8).

$$P(X \geq 10) = 1 - P(X \leq 9) \approx 1 - 0.7166 = \boxed{0.2834}.$$

★ *Intuition:* This additivity is exactly why Poisson is the right model for "total order flow across many independent venues" — you never need to track each venue separately once you just need the aggregate count; the rates simply add, no convolution headache required.

Problem C3 — Poisson thinning (splitting)

Emails arrive at a trading desk as Pois(λ=10) per hour; each is independently "urgent" with probability 0.1. **Thinning theorem:** if each of a Pois(λ) count of events is independently type A w.p. q, then (#type A) $\sim$ Pois(λq). Find P(exactly 2 urgent emails).

Solution. Urgent count $\sim$ Pois(10×0.1) = Pois(1).

$$P(X{=}2) = \frac{e^{-1}1^2}{2!} = \frac{e^{-1}}{2} \approx \boxed{0.1839}.$$

★ *Intuition:* Thinning is the mirror image of C2's additivity: where C2 *merges* independent Poisson streams by adding rates, thinning *splits* one Poisson stream into independent substreams by multiplying the rate by each branch's probability. Together, merging and splitting mean you can freely reorganize Poisson processes in whatever way makes a problem easiest, without ever leaving the Poisson family.

▶ *Speed trick:* Thinning is why "each event independently labeled type A/B/C/..." problems almost always resolve into separate, independent Poissons per label — check for this structure before reaching for a harder joint-distribution approach.

Problem C4 — Rare-event tail & "how often is a bad day?"

A market maker averages $\lambda = 2$ large adverse-selection losses per day (Poisson). Find $P(X \geq 5)$ (a "bad day"), and interpret as "roughly once every how many days?"

Solution.

$$P(X \geq 5) = 1 - P(X \leq 4) \approx 1 - 0.9473 = \boxed{0.0527}.$$

$$\frac{1}{0.0527} \approx \boxed{19} \text{ days.}$$

★ *Intuition:* Converting a small tail probability into "once every N days" ($N = 1/p$) is the natural-frequency trick from Chapter 1.2 again — it turns an abstract 5.3% into something you can actually plan a risk desk around: "expect a bad day roughly every three trading weeks."

D. Uniform Distribution

Problem D1 — PDF, CDF, and a direct probability

Fill-price slippage is modeled as Unif$(-2, 3)$ (basis points). Find the PDF, CDF, $\mathbb{E}[X]$, $\mathrm{Var}(X)$, and $P(-1 < X < 1)$.

Solution. $f(x) = \dfrac{1}{3-(-2)} = \dfrac{1}{5}$ for $x \in [-2, 3]$. $F(x) = \dfrac{x+2}{5}$ on that interval.

$$\mathbb{E}[X] = \frac{-2+3}{2} = \boxed{0.5},$$

$$\mathrm{Var}(X) = \frac{(3-(-2))^2}{12} = \boxed{\tfrac{25}{12}} \approx 2.083.$$

$$P(-1 < X < 1) = F(1) - F(-1) = \tfrac{3}{5} - \tfrac{1}{5} = \boxed{0.4}.$$

★ *Intuition:* Uniform is "maximum ignorance" over a known range — every basis point of slippage in $[-2, 3]$ is equally plausible, which is the natural default model whenever you know the bounds of an outcome but have no reason to favor any value inside them.

Problem D2 — Transformations: the inverse-transform trick

If $X \sim \text{Unif}(0, 1)$, find the distribution of (a) $Y = X^2$ and (b) $Y = -\ln X$.

Solution. **(a)** $F_Y(y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = \sqrt{y}$ for $y \in [0, 1]$, so $f_Y(y) = \dfrac{1}{2\sqrt{y}}$. **(b)** $F_Y(y) = P(-\ln X \leq y) = P(X \geq e^{-y}) = 1 - e^{-y}$ for $y \geq 0$ — this is exactly the CDF of $\boxed{\text{Exp}(1)}$! So $-\ln U \sim \text{Exp}(1)$ whenever $U \sim \text{Unif}(0, 1)$.

★ *Intuition:* Part (b) is the *inverse-transform sampling* method, the single most important trick for turning a computer's uniform random-number generator into *any* other distribution: feed $U \sim \text{Unif}(0,1)$ into the inverse CDF of your target distribution, and out comes a sample from that distribution. Every Monte Carlo simulation you'll ever run leans on this fact.

Problem D3 — Geometric probability: two uniforms and a triangle

$X, Y \overset{\text{iid}}{\sim} \text{Unif}(0,1)$. Find $P(X+Y<1)$ and $\mathbb{E}[X+Y \mid X+Y<1]$.

Solution. $P(X+Y<1)$ is the area of the triangle below the line $x+y=1$ inside the unit square: $\boxed{\frac{1}{2}}$. The density of $S=X+Y$ is triangular: $f_S(s)=s$ for $s\in[0,1]$. Conditioning on $S<1$: $f_{S|S<1}(s)=\dfrac{s}{1/2}=2s$.

$$\mathbb{E}[S \mid S<1]=\int_0^1 s(2s)\,ds=\frac{2}{3}\approx\boxed{0.667}.$$

★ *Intuition:* Geometric probability turns an integral into an area — drawing the unit square and shading the event region is almost always faster than setting up the double integral directly, and it makes the conditional-density computation (which can look intimidating in symbols) visually obvious: you're just re-normalizing the triangular "wedge" of density that remains after conditioning.

E. Exponential Distribution

Problem E1 — Memorylessness, concretely

A server's time-to-next-crash is Exp(λ) with mean $1/\lambda = 30$ days. Given it has run crash-free for 10 days, find P(crash-free for 15 more days).

Solution. By memorylessness, the past 10 crash-free days are irrelevant:

$$P(X>25 \mid X>10)=P(X>15)=e^{-15/30}$$

$$=e^{-0.5}\approx\boxed{0.6065}.$$

★ *Intuition:* The server doesn't "age" — an Exponential clock has no concept of wear-and-tear, only of "still hasn't happened yet." This is the *defining* property of the Exponential (in fact, it's the *only* continuous distribution with this property), and it's why Exponential is the natural model for failures driven by sudden external shocks rather than gradual degradation.

Problem E2 — Racing exponentials: the minimum

Three market makers' quote-update times are independent Exp(λ_1=2), Exp(λ_2=3), Exp(λ_3=5) (rates/sec). Find the distribution of the time until the *first* update, and P(MM 2 updates first).

Solution.

$$\min(X_1,X_2,X_3)\sim\text{Exp}(\lambda_1+\lambda_2+\lambda_3)=\text{Exp}(10).$$

$$P(\text{MM 2 first})=\frac{\lambda_2}{\lambda_1+\lambda_2+\lambda_3}=\frac{3}{10}=\boxed{0.3}.$$

★ *Intuition:* Think of three independent alarm clocks racing to go off first: the combined "something happened" rate is just the sum of the individual rates (faster clocks make the first-ring come sooner, in exactly the additive way you'd guess), and the probability any *specific* clock wins the race is simply its share of the total ringing rate — no integration required.

Problem E3 — Racing to the *last* finisher: $\mathbb{E}$[max] via harmonic numbers

Three i.i.d. components have lifetimes Exp(λ=0.1) (mean 10 days). Find $\mathbb{E}$[min] and $\mathbb{E}$[max] (time until *all three* have failed).

Solution. $\mathbb{E}[\min]=\dfrac{1}{3\lambda}=\dfrac{1}{0.3}\approx\boxed{3.33}$ days. **For the max**, use memorylessness recursively: after the 1st failure (expected time $\frac{1}{3\lambda}$), the other two "restart" by memorylessness, racing at combined rate 2λ (expected additional time $\frac{1}{2\lambda}$); after the 2nd failure, the last one remaining needs expected time $\frac{1}{\lambda}$. Summing:

$$\mathbb{E}[\max]=\frac{1}{3\lambda}+\frac{1}{2\lambda}+\frac{1}{\lambda}=\frac{1}{\lambda}\left(1+\frac{1}{2}+\frac{1}{3}\right)$$

$$=\frac{1}{0.1}\cdot\frac{11}{6}\approx\boxed{18.33}\text{ days.}$$

★ *Intuition:* That $\left(1+\frac{1}{2}+\frac{1}{3}\right)=H_3$ is the exact same harmonic number from the Coupon Collector problem in Notebook 1 — both problems are secretly the same shape: "the first thing to happen is fast, the next thing (racing fewer remaining competitors) is slower, the last holdout is slowest of all," and summing those progressively-slower expected wait times always produces a harmonic series. Once you see this shape, $\mathbb{E}$[max] of n i.i.d. Exponentials is just H_n/λ — no separate derivation needed.

F. Normal Distribution

Problem F1 — Standardization

Daily P&L is $\mathcal{N}(\mu$=\$50k$, \sigma$=\$20k). Find P(loss day) and P(P&L > \$90k).

Solution.

$$Z=\frac{0-50}{20}=-2.5\implies P(X<0)=\Phi(-2.5)\approx\boxed{0.0062}.$$

$$Z=\frac{90-50}{20}=2.0\implies P(X>90)=1-\Phi(2)\approx\boxed{0.0228}.$$

★ *Intuition:* Standardizing is just a change of units — converting "dollars away from average" into "standard deviations away from average" lets you reuse one universal Z-table for every Normal distribution that will ever exist, instead of needing a different table for every (μ,σ) pair.

Problem F2 — The empirical rule, and where it breaks down

Using the 68–95–99.7 rule, approximate $P(|Z|>2)$ and $P(|Z|>3)$. A trader calls a move "a 4-sigma event, once every 15,000 days." Compute the *exact* two-sided tail probability at $Z=4$ and check the claim.

Solution. Empirical rule: $P(|Z|>2)\approx\boxed{5\%}$, $P(|Z|>3)\approx\boxed{0.3\%}$. Exact at $Z=4$: $2(1-\Phi(4))\approx 2(3.167\times10^{-5})=6.334\times10^{-5}$.

$$\frac{1}{6.334\times10^{-5}}\approx\boxed{15{,}787}\text{ days.}$$

The trader's "once every 15,000 days" is remarkably close to the exact Normal-theory value!

★ *Intuition:* This is a double-edged lesson. On one hand, the exact math backs up the "1-in-15,000" trading-desk folklore almost perfectly *if returns were truly Normal.* On the other hand, this is exactly the gap that gets real trading desks in trouble: real market returns have *fatter tails* than the Normal distribution, so a move that "should" happen once every 15,000 days under a Normal model can happen far more often in reality. The 68–95–99.7 rule is a good quick mental anchor out

to about 2σ; beyond 3σ, always compute the exact tail rather than linearly extrapolating the rule, and remember the exact tail itself may understate real-world risk.

Problem F3 — Linear combinations of independent Gaussians

Strategy A $\sim \mathcal{N}(2\%, (5\%)^2)$, Strategy B$\sim \mathcal{N}(1\%, (3\%)^2)$, independent. A portfolio holds 60% A, 40% B. Find the portfolio return's distribution and $P(\text{portfolio return} < 0)$.

Solution.

$$\mu_Y = 0.6(2\%) + 0.4(1\%) = \boxed{1.6\%}.$$

$$\sigma_Y^2 = 0.6^2(5\%)^2 + 0.4^2(3\%)^2 = 9 + 1.44 = 10.44\%^2$$

$$\implies \sigma_Y \approx \boxed{3.23\%}.$$

$$Z = \frac{0 - 1.6}{3.23} \approx -0.495 \implies P(Y < 0) \approx \boxed{0.310}\ (31.0\%).$$

★ *Intuition:* Note the variance does *not* shrink simply because you weighted two strategies together — it's a weighted sum of *squared* weights times variances, not a simple weighted average of volatilities. This is the mathematical seed of portfolio diversification: combining independent (uncorrelated) return streams reduces relative risk per unit of expected return, but you must always combine through variances (and later, covariances), never by naively averaging standard deviations.

▶ *Speed trick:* A linear combination of independent Normals is always Normal — never attempt to track the combination's exact pdf by convolution by hand; just compute the new mean and variance using the two formulas above and standardize as usual.

Drill order for review: re-derive A3 (mode tie), B3 (decomposition trick), D2(b) (inverse-transform), E3 ($\mathbb{E}[\max]$ via harmonic numbers), and F2 (exact vs. empirical-rule tail) from scratch, closed-book. Together they carry the chapter's five reusable ideas: the mode-tie condition, breaking a hard waiting time into simple i.i.d. pieces, generating any distribution from a Uniform, the memorylessness-recursion trick for order statistics of Exponentials, and the limits of the empirical rule.